

Paper Completion Checklist

Representations of Lie Groups: A Mathematical Dissection of Quantum Spin

Generated July 20, 2026

10-Day Sprint: Jul 21–31 (Presentation on Fri 31) — Jul 20 lost to room move										
21	22	23	24	25	26	27	28	29	30	31
Tue	Wed	Thu	Fri	Sat	Sun	Mon	Tue	Wed	Thu	Fri
D1	D2	D3	D4	D5	D6	D7	D8	D9	Buffer/Rev.	PRES.

Tue 21 Jul — Day 1: $\mathfrak{so}(3)$ & $\mathfrak{su}(2)$ Lie Algebras

- ☐ Define $\mathfrak{so}(3) = T_{\mathbf{I}_3}\mathrm{SO}(3) = \{\mathbf{R} \in \mathbb{R}^{3,3} \mid \mathbf{R}^\top = -\mathbf{R}\}$
- ☐ Define basis $\mathbf{L}_1, \mathbf{L}_2, \mathbf{L}_3$ with explicit matrices
- ☐ Prove $[\mathbf{L}_i, \mathbf{L}_j] = \varepsilon_{ijk}\mathbf{L}_k$ and $\dim \mathfrak{so}(3) = 3$
- ☐ Define $\mathfrak{su}(2) = T_{\mathbf{I}_2}\mathrm{SU}(2) = \{\mathbf{U} \in \mathbb{C}^{2,2} \mid \mathbf{U}^\dagger = -\mathbf{U}, \mathrm{tr}(\mathbf{U}) = 0\}$
- ☐ Define basis $\mathbf{T}_i = \frac{1}{2i}\sigma_i$ with explicit Pauli matrices
- ☐ Prove $[\mathbf{T}_i, \mathbf{T}_j] = \varepsilon_{ijk}\mathbf{T}_k$ and $\dim \mathfrak{su}(2) = 3$

Wed 22 Jul — Day 2: Double Cover & Lie Algebra Isomorphism

- ☐ Define $\Phi : \mathrm{SU}(2) \rightarrow \mathrm{SO}(3)$ by $\Phi(\mathbf{U})(\mathbf{X}) = \mathbf{U}\mathbf{X}\mathbf{U}^\dagger$
- ☐ Prove $\Phi(\mathbf{U}) \in \mathrm{SO}(3)$ (preserves inner product and determinant)
- ☐ Prove Φ is a smooth group homomorphism
- ☐ Prove Φ is surjective with $\ker \Phi = \{\pm \mathbf{I}_2\}$ (2-to-1)
- ☐ Conclude $\mathrm{SU}(2)$ is the universal double cover of $\mathrm{SO}(3)$
- ☐ Define $\phi : \mathfrak{so}(3) \rightarrow \mathfrak{su}(2)$ by $\phi(\mathbf{L}_i) = \mathbf{T}_i$
- ☐ Prove ϕ is a Lie algebra isomorphism (linear + preserves bracket)

Thu 23 Jul — Day 3: Complexification & Ladder Operators

- ☐ Define complexification: $\mathfrak{g}_{\mathbb{C}} = \mathfrak{g} \otimes_{\mathbb{R}} \mathbb{C}$ for a real Lie algebra $\mathfrak{g}$
- ☐ Define complex representation of a real Lie algebra (extends to $\mathfrak{g}_{\mathbb{C}}$)
- ☐ Prove $\mathfrak{su}(2)_{\mathbb{C}} \cong \mathfrak{sl}(2, \mathbb{C}) = \{\mathbf{A} \in \mathbb{C}^{2,2} \mid \mathrm{tr}(\mathbf{A}) = 0\}$
- ☐ Define Cartan–Weyl basis: $\mathbf{H} = 2i\mathbf{T}_3, \mathbf{E} = i\mathbf{T}_1 - \mathbf{T}_2, \mathbf{F} = i\mathbf{T}_1 + \mathbf{T}_2$
- ☐ Write explicit matrix forms of $\mathbf{H}, \mathbf{E}, \mathbf{F}$
- ☐ Prove commutation relations: $[\mathbf{H}, \mathbf{E}] = 2\mathbf{E}, [\mathbf{H}, \mathbf{F}] = -2\mathbf{F}, [\mathbf{E}, \mathbf{F}] = \mathbf{H}$

Fri 24 – Sat 25 Jul — Days 4–5: Classification of Irreducible Representations

- ☐ State the theorem: for each $j \in \{0, \frac{1}{2}, 1, \frac{3}{2}, \dots\}$, $\exists!$ irrep $\pi_j : \mathfrak{su}(2) \rightarrow \mathfrak{gl}(V_j)$ with $\dim V_j = 2j+1$
- ☐ Define weight spaces and prove $\pi_j(\mathbf{H})$ diagonalizes on V_j
- ☐ Ladder Lemma: $\pi_j(\mathbf{E})$ raises weight by $+2$, $\pi_j(\mathbf{F})$ lowers by -2

- ☐ Highest weight argument: pick $\mathbf{v} \neq \mathbf{0}$ with $\pi_j(\mathbf{E})\mathbf{v} = \mathbf{0}$, let λ be its $\mathbf{H}$ -eigenvalue
- ☐ Construct basis $\mathbf{v}_k = \pi_j(\mathbf{F})^k \mathbf{v}$ and prove $\pi_j(\mathbf{E})\mathbf{v}_k = k(\lambda - k + 1)\mathbf{v}_{k-1}$
- ☐ Prove $\lambda = 2j \in \mathbb{N}_0$ and the chain terminates at $\mathbf{v}_{2j+1} = \mathbf{0}$
- ☐ Conclude $\dim V_j = 2j + 1$ and weights are $\{-2j, -2j + 2, \dots, 2j - 2, 2j\}$
- ☐ Normalize to get the standard ladder formulas with square-root factors
- ☐ Prove uniqueness: any irrep of dimension $2j + 1$ is equivalent to π_j
- ☐ Classification table: $j = 0$ scalar, $\frac{1}{2}$ spinor, 1 vector, $\frac{3}{2}, \dots$

Sun 26 Jul — Day 6: Integration to $\mathrm{SU}(2)$

- ☐ Prove $\mathrm{SU}(2) \cong \mathbb{S}^3$ (homeomorphic to the 3-sphere)
- ☐ Conclude $\mathrm{SU}(2)$ is simply connected
- ☐ Theorem: every Lie algebra rep $\pi : \mathfrak{su}(2) \rightarrow \mathfrak{gl}(V)$ integrates uniquely to a Lie group rep $\Pi : \mathrm{SU}(2) \rightarrow \mathrm{GL}(V)$ via $\Pi(e^{\mathbf{X}}) = e^{\pi(\mathbf{X})}$
- ☐ Apply to π_j to get $\Pi_j : \mathrm{SU}(2) \rightarrow \mathrm{GL}(V_j)$

Mon 27 Jul — Day 7: Projective Representations & Descent

- ☐ Define projective representation: $\rho(g)\rho(h) = \omega(g, h)\rho(gh)$
- ☐ Bargmann's theorem / physical motivation: in QM, symmetries need only be represented up to phase
- ☐ Prove: every projective representation of $\mathrm{SO}(3)$ lifts to an ordinary representation of $\mathrm{SU}(2)$
- ☐ Prove descent criterion: $\Pi_j(-\mathbf{I}_2) = e^{2\pi \cdot \pi_j(\mathbf{T}_3)} = (-1)^{2j} I_{V_j}$
- ☐ Conclude: integer j descends to ordinary $\mathrm{SO}(3)$ rep; half-integer j gives projective $\mathrm{SO}(3)$ rep (spinor)

Tue 28 Jul — Day 8: Physical Spin Operators

- ☐ Define Hermitian spin observables: $S_k = i\hbar \pi_j(\mathbf{T}_k)$
- ☐ Prove angular momentum algebra: $[S_i, S_j] = i\hbar \varepsilon_{ijk} S_k$
- ☐ Prove $S^2 = \hbar^2 j(j + 1)I$ and S_z has eigenvalues $\hbar m$ ($m = -j, \dots, j$)
- ☐ Example: $j = \frac{1}{2}$ — Pauli matrices $S_k = \frac{\hbar}{2} \sigma_k$ with explicit matrix forms
- ☐ Spinor sign change: rotation by 2π maps $|\psi\rangle \mapsto -|\psi\rangle$; restored after 4π

Wed 29 Jul — Day 9: Polish & Submit

- ☐ Proof-read all new content for mathematical correctness
- ☐ Ensure consistent notation with earlier sections
- ☐ Add cross-references (`\ref`) throughout new sections
- ☐ Run full compile with `latexmk` — zero errors, zero warnings
- ☐ Update abstract if needed to reflect completed content
- ☐ Final `./publish` to push to GitHub

Thu 30 Jul — Buffer & Presentation Prep

- ☐ Catch up on any unfinished items from Days 1–9

- ☐ Prepare presentation slides / talk outline
 - ☐ Final paper read-through on paper
 - ☐ Rehearse presentation with poster
 - ☐ Verify QR code link works, PDFs are accessible on GitHub
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Fri 31 Jul — PRESENTATION DAY

Feasibility: 9 working days, 1 buffer/prep day. Still on track.

- Losing Mon 20 Jul to a room move costs the extra buffer day, but the schedule remains realistic.
 - Days 1–3 (Tue–Thu) are straightforward — **easy**.
 - Days 4–5 (Fri–Sat) classification theorem, the hardest part, now falls on a **weekend** — good.
 - Day 6 (Sun) integration is light — **recovery day** if classification spills over.
 - Days 7–9 (Mon–Wed) are moderate polish — standard Lie theory + cleanup.
 - Thu 30 Jul is the only buffer/prep day — **use it wisely**. If you stay on schedule, it's a full presentation prep day.
 - **Total: ~17 new items.** Tight but well within your demonstrated writing speed.
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